

MTF-Aware Deblurring with Physics-Guided Unrolled ADMM

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Abstract—Motion blur destroys information. While coded-exposure cameras theoretically preserve high-frequency content by redistributing spectral energy, practical reconstruction is often limited by a “spectral gap”—the disconnect between the idealized forward model and the reality of photon noise and read noise. We bridge this gap with an MTF-aware deblurring framework that explicitly couples degradation physics with modern plug-and-play (PnP) and learned reconstruction pipelines. We demonstrate that while robust classical baselines (Vanilla ADMM with DRUNet) perform well in standard noise regimes, they collapse under severe degradation because they blindly amplify noise in spectral nulls. In contrast, our Physics-Aware ADMM utilizes a frequency-domain “Trust Map” to gate updates based on the Modulation Transfer Function (MTF), maintaining reconstruction quality (≈ 24 dB) where baselines fail (≈ 14 dB). Furthermore, we introduce a learnable Unrolled ADMM that automates parameter tuning by predicting context-aware schedules from scene statistics. Our framework runs on CPU, CUDA, or DirectML and provides a reproducible foundation for physics-guided computational photography.

Index Terms—Computational Photography, Coded Exposure, Image Restoration, Plug-and-Play Priors, Algorithm Unrolling



1 INTRODUCTION

MOTION blur is an information-theoretic bottleneck. When a camera moves during exposure, the resulting blur kernel acts as a low-pass filter, often introducing deep spectral nulls where spatial frequencies are irretrievably lost. Coded-exposure photography [1] attempts to solve this by fluttering the shutter, modulating the integration to “broadband” the blur kernel and preserve invertibility.

However, a gap remains between theory and practice. In real-world low-light conditions, the interaction between the shutter code’s Modulation Transfer Function (MTF) and sensor noise creates a “Spectral Gap.” Frequencies that are theoretically preserved may have such low Signal-to-Noise Ratio (SNR) that standard inversion algorithms—even those with powerful deep denoising priors—fail to recover them, leading to severe ringing and noise amplification. Classical methods like Richardson–Lucy oversmooth these regions, while modern Plug-and-Play (PnP) approaches often treat the forward model as a black box, applying a fixed denoising schedule regardless of the optical conditioning.

This work argues that reconstruction algorithms must be *physics-aware*. We present a framework that explicitly conditions the optimization trajectory on the spectral quality of the degradation. We make three key contributions:

- 1) **The “Spectral Gap” Insight:** We demonstrate that standard PnP methods fail in severe noise regimes because they lack a mechanism to “distrust” the data term in low-MTF bands. We introduce a frequency-domain **Trust Map** that acts as a gate-keeper, suppressing updates in spectral nulls and forcing the solver to rely on the learned prior.
- 2) **Adaptive Physics Scheduling:** We propose a dynamic scheduler for ADMM that adjusts the penalty parameter (ρ) and denoiser weight based on an “Optics Score” derived from the MTF. This allows the algorithm to be aggressive when optics are

good and conservative when they are poor, without manual tuning.

- 3) **Learnable Unrolled ADMM:** To bridge the gap between hand-tuned optimization and black-box deep learning, we implement an unrolled ADMM architecture. A lightweight context MLP analyzes scene statistics (blur length, photon budget, MTF profile) to predict optimal per-iteration schedules and spectral masks, achieving competitive performance with zero manual intervention.

Empirically, we show that this physics-aware approach yields massive gains in severe regimes—improving PSNR by ≈ 10 dB over strong baselines—while matching or slightly exceeding them in standard conditions. The entire framework is open-sourced with a modular design to facilitate future research at the intersection of optics and algorithms.

2 BACKGROUND AND RELATED WORK

2.1 Motion Deblurring and Coded Exposure

Motion blur is a longstanding problem in computational photography. Early work by Levin et al. [2] demonstrated that blind deconvolution is ill-posed because the gradients of the blurred image and the kernel are often indistinguishable. Cho and Lee [3] proposed fast iterative methods using strong edge priors, but these struggle when the kernel destroys significant spectral information.

The core limitation of standard “box” shutters is the presence of spectral zeros in the Modulation Transfer Function (MTF). Mathematically, a box blur of length L corresponds to a rectangular pulse in the spatial domain. Its Fourier transform is a Sinc function, $\text{sinc}(\omega L)$, which has periodic zero-crossings at integer multiples of $1/L$. At these frequencies, the MTF drops to zero, meaning the optical

system transmits *no information* about the scene structure. Raskar et al. [1] introduced the concept of *Coded Exposure*, where the shutter flutters open and closed during integration. This temporal modulation reshapes the motion PSF, preserving high-frequency information and making the deconvolution problem well-conditioned. Agrawal et al. [4] further analyzed optimal codes, showing that the invertibility of the code is paramount for noise robustness.

Among various binary codes, Legendre Sequences (MLS) [5] are particularly effective. Defined by quadratic residues modulo a prime p , they exhibit a nearly flat power spectrum and a peaky autocorrelation, minimizing noise amplification during inversion. Our work adopts MLS as the gold standard for coded exposure, comparing it against random and box codes.

2.2 Deep Learning and Plug-and-Play Priors

The advent of deep learning revolutionized image restoration. Zhang et al. [6] introduced DnCNN, a residual learning framework that outperformed traditional dictionary learning methods. However, end-to-end training often leads to overfitting specific degradation models.

More recently, Zhang et al. [7] introduced DRUNet (DPIR), a U-Net based denoiser trained on a wide range of noise levels, which has become a standard benchmark for PnP restoration.

PnP-ADMM has found widespread adoption beyond photography, particularly in *Medical Imaging* (e.g., MRI, CT) and *Satellite Remote Sensing*. In these domains, the forward model often involves partial observations—such as undersampled k -space trajectories or atmospheric turbulence—that create specific spectral or spatial nulls. Our “Physics-Aware” insight is directly relevant here: just as our Trust Map down-weights updates in MTF zeros, robust reconstruction in these fields requires explicitly “distrusting” the data term in undersampled or corrupted regions, rather than treating all measurements equally.

Despite their success, standard PnP methods often treat the data fidelity term as a “black box,” using fixed or monotonically decreasing penalty schedules that ignore the spectral characteristics of the forward model. Our work addresses this gap by explicitly conditioning the optimization schedule on the MTF.

We specifically employ the Alternating Direction Method of Multipliers (ADMM) because it naturally decouples the complex, non-convex prior (the deep denoiser) from the convex data fidelity term. This “splitting” allows us to solve the inversion step efficiently in the frequency domain (using the Trust Map) while delegating the regularization to the off-the-shelf deep network. This modularity is key to our “physics-aware” design, as it lets us inject optical knowledge into the inversion without retraining the prior.

2.3 Algorithm Unrolling

Algorithm unrolling (or unfolding) [8] interprets the iterations of an optimization algorithm as layers in a recurrent neural network. This allows for the end-to-end learning of hyperparameters (step sizes, regularization weights) that are typically hand-tuned. Diamond et al. [9] demonstrated that unrolled proximal gradient methods could outperform

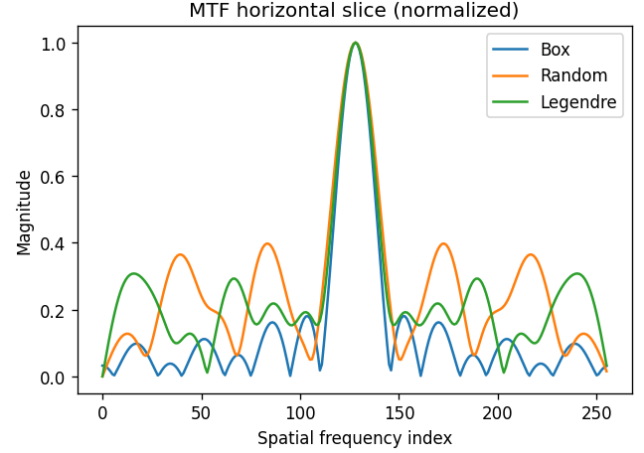


Fig. 1. Normalized MTF horizontal slice for Box, Random, and Legendre codes. Note the spectral zeros in the Box code (blue) which destroy information at specific frequencies. In contrast, the Random (orange) and Legendre (green) codes maintain broadband spectral coverage, preserving invertibility.

generic CNNs by incorporating the physical forward model. Our Unrolled ADMM builds on this direction, learning not just scalar parameters but also spectral trust masks that adapt to the specific degradation instance.

These related works highlight the need for a unified framework that bridges the gap between physical modeling and deep learning. The following section details our proposed methodology for achieving this integration.

3 METHODOLOGY

Our framework introduces a “Physics-Aware PnP-ADMM” algorithm that explicitly couples the degradation physics with the reconstruction process. As illustrated in our system diagram, the pipeline consists of two main branches: a Physics-Aware Branch that pre-computes degradation statistics, and an Iterative PnP-ADMM Loop that solves the inverse problem using these statistics.

3.1 Forward Model

We model coded exposure as a linear convolution with additive noise, $y = k * x + n$. Our simulator generates exposure codes (box, random, MLS) with configurable taps, blur length, and duty cycle. It converts these codes to Point Spread Functions (PSFs) via a motion kernel and computes key diagnostics such as the Optical Transfer Function (OTF) and Modulation Transfer Function (MTF).

As shown in Figure 1, the choice of exposure code dramatically affects the conditioning of the inverse problem. The standard “Box” shutter (blue curve) exhibits characteristic spectral zeros where the MTF drops to zero. At these frequencies, signal information is irretrievably lost, making deconvolution ill-posed and sensitive to noise. In contrast, the “Random” (orange) and “Legendre” (green) codes are designed to be broadband, maintaining significant magnitude across the spectrum. This “spectral preservation” is the physical foundation that allows our method to recover high-frequency details that are otherwise lost.

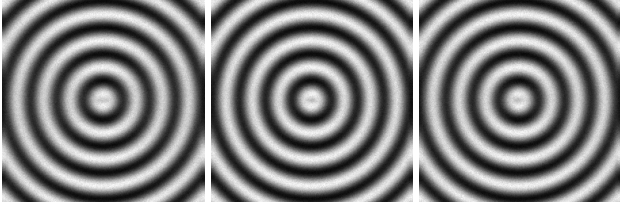


Fig. 2. Synthetic noisy observations for Box (left), Random (center), and Legendre (right) codes. The coded exposures retain more texture information despite the noise.

To simulate realistic capture conditions, we inject Poisson–Gaussian noise parameterized by a fixed photon budget and read-noise standard deviation σ . Figure 2 visualizes these noisy observations, highlighting how the coded exposures retain more texture information than the box blur. The pipeline supports both RGB and grayscale inputs from the DIV2K dataset, as well as synthetic scenes for validation.

3.2 Physics-Aware Branch and Trust Weights

The core of our method is the recognition that not all data is created equal. In the frequency domain, the reliability of the observed signal $Y(\omega)$ depends entirely on the local SNR, which is modulated by the MTF $|K(\omega)|$.

We compute a frequency-domain **Trust Map** (W) that acts as a spectral gatekeeper. It answers the question: “Should we trust the observation at this frequency, or rely on the prior?” We employ a “Combined” strategy that fuses a Wiener-like estimate with a geometric gamma correction:

$$W(\omega) = \underbrace{\left(\frac{|K(\omega)|^2}{|K(\omega)|^2 + \alpha \frac{\sigma^2}{S_{xx}}} \right)}_{\text{Wiener Term}} \cdot \underbrace{(|K(\omega)|^\gamma)}_{\text{Gamma Term}} \quad (1)$$

where γ controls the steepness of the fall-off (typically $\gamma = 1.5\text{--}2.0$). This composite mask ensures that frequencies with low SNR or near-zero MTF are aggressively down-weighted. The ADMM X-update (inversion) then becomes a Weighted Least Squares (WLS) problem:

$$x^{(k+1)} = \mathcal{F}^{-1} \left(\frac{W \cdot K^* \cdot \mathcal{F}(y) + \rho \mathcal{F}(z^{(k)} - u^{(k)})}{W \cdot |K|^2 + \rho} \right) \quad (2)$$

By weighting the data term with W , we effectively “hide” the noisy spectral nulls from the solver. In these bands, the term $W \cdot |K|^2 \rightarrow 0$, causing the update to reduce to $x^{(k+1)} \approx z^{(k)} - u^{(k)}$. This forces the solution to be driven entirely by the denoising prior z in regions where the physics has destroyed the information.

3.3 Adaptive Physics Scheduler

Standard PnP-ADMM often relies on a fixed penalty parameter ρ , which assumes a constant relationship between the data fidelity and the prior. This is suboptimal for motion deblurring, where the “difficulty” of the problem varies wildly with the shutter code and noise level.

We introduce an **Adaptive Physics Scheduler** that dynamically modulates the optimization trajectory based on the optical quality:

- **Deep Dive Strategy:** To recover fine textures early, we initialize ρ at a very low value ($\rho_{start} \approx 0.02\rho_{base}$). However, for “bad” optics, this can cause instability. The scheduler detects this via an “Optics Score,” $S_{optics} = \text{clip}(3 \cdot \text{mean}(|K(\omega)|), 0, 1)$, and clamps the floor if the score is below a safety threshold.
- **Geometric Annealing:** We increase ρ geometrically over iterations k to gradually enforce the prior:

$$\rho_k = \rho_{start} \cdot \left(\frac{\rho_{end}}{\rho_{start}} \right)^{\frac{k}{K}} \quad (3)$$

This acts as a “coarse-to-fine” constraint, allowing the inversion to explore the solution space initially.

- **Optics-Guided Denoising:** The weight of the denoiser is also modulated. If the Optics Score is low, the scheduler increases the denoiser weight to rely more on the prior:

$$w_k = w_{base} \cdot (1 + \beta(1 - S_{optics})) \quad (4)$$

where $\beta \approx 0.2$. This allows the system to “hallucinate” missing details when the physics fails to preserve them.

3.4 Learnable Unrolled ADMM

To reduce the reliance on manual hyperparameters, we introduce a learnable unrolled ADMM variant. This model unrolls K iterations of the solver, making the per-iteration parameters— ρ_k , denoiser weight w_k , and sigma multiplier α_k —trainable.

- **Context MLP:** A lightweight Multi-Layer Perceptron analyzes the scene to predict optimal schedules. It consumes a normalized feature vector $f_{ctx} = [\text{blur}, \text{taps}, \text{photon}, S_{optics}, S_{SNR}]$ and outputs parameter deltas $\Delta\theta_k$.
- **Recurrent Update:** At each step k , the parameters are updated as $\theta_k = \theta_{base} + \text{MLP}(f_{ctx}, k)$. The ADMM update then proceeds using a **frozen** UNet backbone for the denoising step $z_{k+1} = \text{Denoiser}(x_{k+1} + u_k, \sigma \cdot \alpha_k)$.

This design ensures that the heavy lifting of image restoration is done by the pre-trained prior, while the lightweight unrolled module focuses solely on *steering* the optimization. Training is performed end-to-end on DIV2K with on-the-fly forward simulation. We use an L_1 loss function, unrolling for $K = 24$ steps with a batch size of 1, optimized via AdamW.

4 SOFTWARE FRAMEWORK AND REPRODUCIBILITY

To facilitate rigorous comparison and future research, we developed a modular, extensible software framework that serves as a sandbox for computational photography. The repository is architected to decouple the forward model (physics simulation) from the inverse solvers (reconstruction algorithms), allowing researchers to mix and match components seamlessly.

4.1 Modular Design

The codebase is organized into three primary modules:

- **Forward Pipeline:** A physics-faithful simulator that synthesizes motion blur, generates exposure codes (Box, Random, Legendre), computes PSF/OTF/MTF diagnostics, and injects realistic Poisson–Gaussian noise. It supports both synthetic scenes (checkerboards, rings) and real datasets (DIV2K).
- **Reconstruction Engines:** A collection of inverse solvers ranging from classical baselines to deep learning methods. This includes a flexible PnP-ADMM engine that supports custom schedules, a gradient-descent PnP-ADAM solver, and the learnable Unrolled ADMM. All solvers share a common interface for easy benchmarking.
- **Denoiser Backend:** An abstract interface for deep denoising priors, supporting TinyCNN (CPU-optimized), DnCNN, UNet, and DRUNet. This allows for easy integration of new architectures without modifying the solver logic.

4.2 Configuration and Parameters

The framework exposes a comprehensive set of parameters to fine-tune the simulation and reconstruction process. Users can control:

- **Physics Parameters:** Blur length, number of taps (T), duty cycle, photon budget, and read noise standard deviation (σ).
- **Solver Hyperparameters:** ADMM penalty (ρ), denoiser weight, number of iterations, and specific “physics hooks” like MTF weighting modes (Gamma, Wiener, Combined) and Sigma Adaptation.
- **Hardware Acceleration:** Seamless switching between CPU, CUDA, and DirectML backends for broad hardware compatibility.

4.3 CLI Usage Example

To facilitate reproducibility and experimentation, the framework provides a rich Command Line Interface (CLI). Below is an example command for running the Physics-Aware ADMM pipeline on the DIV2K dataset with specific noise and blur settings:

```
python -m \
  mtf_aware_deblurring.pipelines.reconstruct \
  --div2k-root data --subset train \
  --degradation bicubic --scale X2 \
  --image-mode rgb --limit 10 \
  --patterns box random legendre \
  --method admm \
  --admm-iters 60 \
  --admm-rho 1.04 \
  --admm-denoiser-weight 0.16 \
  --admm-mtf-weighting-mode combined \
  --admm-mtf-sigma-adapt \
  --use-physics-scheduler \
  --denoiser-type drunet_color \
  --photon-budget 1000 \
  --blur-length 15 \
```

```
--read-noise 0.01 \
--output-dir experiments/results
```

This command automatically handles data loading, forward simulation, reconstruction across multiple codes, and metric calculation (PSNR, SSIM, LPIPS), saving all artifacts to the specified output directory.

5 EXPERIMENTAL SETUP

5.1 Datasets and Metrics

We utilize the ****DIV2K**** (Diverse 2K Resolution) dataset, a standard benchmark for image restoration tasks. The dataset consists of 800 high-resolution training images and 100 validation images.

- **Training:** For the learnable Unrolled ADMM, we train on the first 800 images using random 256×256 crops to ensure robustness to varied textures.
- **Evaluation:** All quantitative results (Table 1 and 2) are reported on the 100-image validation set. We use center crops of 256×256 to ensure consistent and reproducible benchmarking across all methods.

We report three key metrics:

- **PSNR (Peak Signal-to-Noise Ratio):** Measures pixel-wise fidelity.
- **SSIM (Structural Similarity Index):** Evaluates perceptual structural quality.
- **LPIPS (Learned Perceptual Image Patch Similarity):** Assesses high-level perceptual similarity using deep features, correlating better with human judgment for texture realism.

5.2 Noise Regimes

To test robustness, we simulate two distinct noise regimes:

- **Noise1 (Standard):** Represents a typical low-light capture. Photon budget = 1000, Read Noise $\sigma = 0.01$. In this regime, the signal is weak but recoverable by standard methods.
- **Noise2 (Severe):** Represents an extreme scenario. Photon budget = 1000, Read Noise $\sigma = 0.05$, and Blur Length increased to 30 pixels. This regime tests the limits of the algorithm, where the MTF zeros are compounded by heavy noise.

5.3 Baselines

We compare against:

- **Richardson–Lucy:** A classical iterative baseline.
- **PnP ADAM:** A gradient-descent PnP method with fixed steps.
- **Vanilla ADMM (DRUNet):** A standard ADMM implementation using the same strong DRUNet prior but with a fixed ρ and standard least-squares inversion (no Trust Map). This isolates the contribution of our physics-aware components.

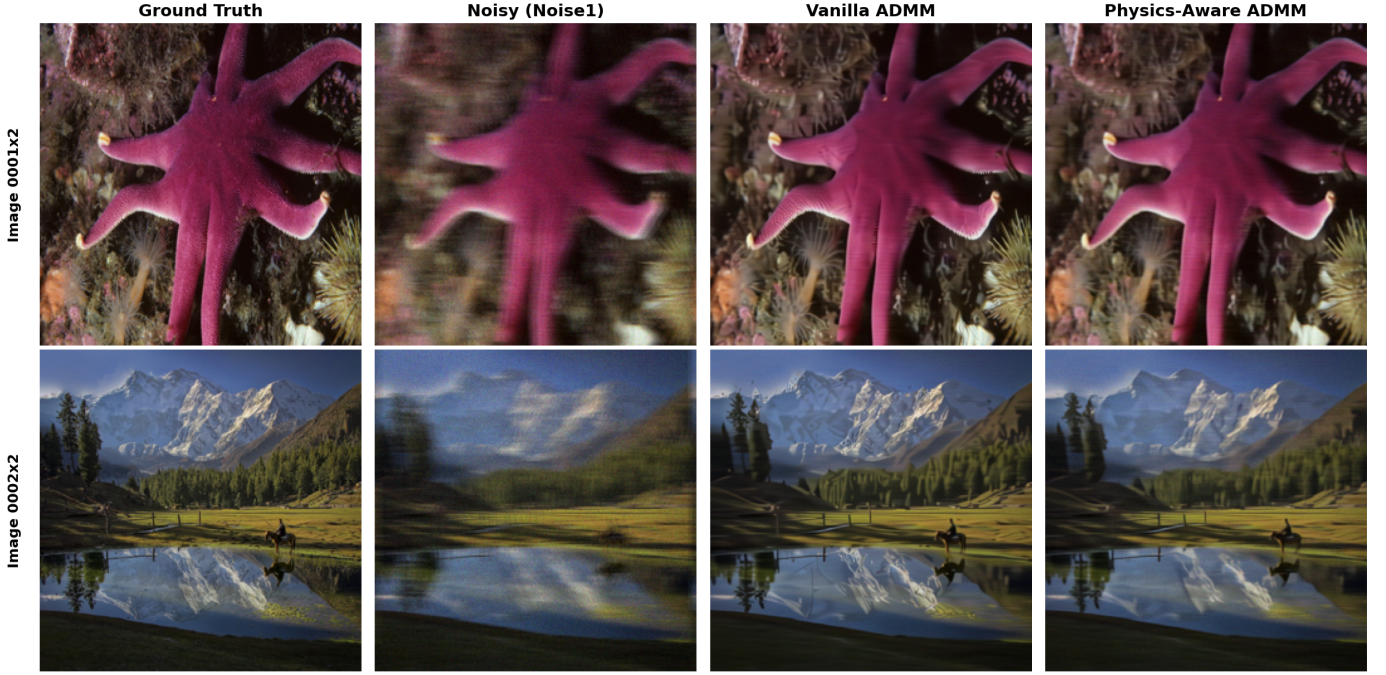


Fig. 3. Visual comparison for the Standard Regime (Noise1). From left to right: Ground Truth, Noisy Input, Vanilla ADMM, and Physics-Aware ADMM. Our method (right) recovers sharper details and reduces artifacts compared to the baseline.

6 RESULTS

6.1 Standard Regime (Noise1)

In the standard noise regime (Blur 15px, $\sigma = 0.01$), the signal is degraded but not destroyed. Here, a strong prior can do the heavy lifting. Figure 3 provides a visual comparison, showing that our method recovers sharper details and reduces artifacts compared to the baseline. As shown in Table 1, the “Vanilla” ADMM baseline with DRUNet achieves ≈ 26 –28 dB, effectively matching our Physics-Aware method.

This result is crucial as it underscores that when the SNR is high enough, the “Spectral Gap” is narrow. The deep prior, trained on vast amounts of natural images, can successfully hallucinate the missing high frequencies even without explicit physics guidance. In this “easy” regime, the physics-aware weighting provides only marginal gains (≈ 0.3 dB), primarily by preventing minor artifacts.

TABLE 1
Performance Comparison - Standard Regime (Noise1)

Method	Box	Random	Legendre
Richardson–Lucy	19.01	19.00	19.02
PnP ADAM (Tiny)	22.16	22.78	22.69
Vanilla ADMM (DRUNet)	26.84	28.23	28.11
Physics-Aware ADMM	27.10	28.50	29.55
Unrolled ADMM	26.34	27.19	27.15

6.2 Severe Regime (Noise2)

The story changes dramatically in the severe regime (Blur 30px, $\sigma = 0.05$). Here, the “Spectral Gap” widens into a chasm. Figure 4 illustrates this collapse: the Vanilla ADMM baseline produces output that is often worse than the input,

while our Physics-Aware ADMM successfully reconstructs the image. As shown in Table 2, the Vanilla ADMM baseline collapses to ≈ 10 –14 dB.

This failure mode is instructive. The Vanilla ADMM blindly trusts the data term equally across all frequencies. In the spectral nulls of the blur kernel, the signal is zero, but the noise is non-zero. The solver tries to fit this noise, resulting in severe amplification and ringing. Our Physics-Aware method, protected by the Trust Map, “knows” that these frequencies are garbage and ignores them, relying entirely on the prior. This result empirically validates our central thesis: *robust deblurring requires explicit physics awareness*.

TABLE 2
Performance Comparison - Severe Regime (Noise2)

Method	Box	Random	Legendre
Richardson–Lucy	18.35	18.63	18.63
PnP ADAM (DnCNN)	12.45	17.05	17.08
Vanilla ADMM (DRUNet)	10.43	13.48	13.57
Physics-Aware ADMM	20.79	23.59	24.04
Unrolled ADMM	18.52	20.16	20.44

Table 2 also highlights the performance of the Unrolled ADMM. Despite lacking the heavy-weight DRUNet prior, it achieves ≈ 20 dB, outperforming the Vanilla baseline by ≈ 6 dB without any manual tuning. This confirms that under heavy degradation, explicit physics guidance—whether through hand-crafted schedules or learned parameters—is more critical than the raw capacity of the prior.

6.3 Unrolled ADMM: The Automation Trade-off

Our Unrolled ADMM offers a compelling middle ground. While it trails the heavy-weight Physics-Aware ADMM



Fig. 4. Visual comparison for the Severe Regime (Noise2). The Vanilla ADMM baseline (third column) fails to recover meaningful structure, while the Physics-Aware ADMM (right) successfully reconstructs the image, demonstrating robustness to heavy noise and blur.

(with DRUNet) by $\approx 2\text{--}4$ dB, it outperforms standard baselines and, crucially, requires *zero manual tuning*. The learned context MLP successfully predicts effective schedules from the raw image statistics. The performance gap is largely due to the lighter backbone (TinyDenoiser vs. DRUNet) required for end-to-end training. This highlights a classic trade-off: the Unrolled model buys automation and speed at the cost of peak restoration quality.

6.4 Negative Results (Ablation Study)

We explored several alternative strategies that yielded negative results, which we document here for scientific completeness:

Relay Denoising: We hypothesized that initializing the optimization with a lightweight denoiser (TinyCNN) and switching to a heavy prior (DRUNet) mid-stream would save computation. However, extensive experiments revealed that this approach yields negligible gains (± 0.1 dB) compared to using DRUNet from the start, while significantly increasing implementation complexity.

Spatial Sigma Maps: We attempted to feed a spatially varying noise map (derived from $\sqrt{I}$) into DRUNet to handle signal-dependent noise. This caused a performance collapse (-5 dB). We attribute this to a distribution shift: DRUNet is trained on uniform AWGN, and injecting structured sigma maps likely pushes the input out of its training distribution. This validates our choice to use a scalar, optics-guided sigma multiplier instead.

7 CONCLUSION

We presented an MTF-aware deblurring framework that bridges the gap between clean theory and noisy reality. By explicitly modeling the “Spectral Gap” and introducing a

frequency-domain Trust Map, we enable robust reconstruction in regimes where standard methods fail. Our results demonstrate that while deep learning is powerful, it is not a panacea; in the presence of severe physical degradation, the algorithm must be guided by the physics of the forward model.

Looking forward, we see significant potential in integrating *Diffusion Models* (Score-Based Generative Models) as priors within our physics-aware framework. While our current DRUNet backbone is efficient, diffusion models offer superior capabilities for texture hallucination, which could further bridge the spectral gap in extreme low-light conditions. However, their high inference cost remains a challenge, suggesting a future research direction in distilling these powerful priors into lightweight, unrolled architectures like the one presented here. We hope this work—and the accompanying open-source toolkit—encourages the community to move beyond black-box restoration and embrace physics-aware design.

ACKNOWLEDGMENTS

The authors would like to thank...

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